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A single-phase five-level multilevel inverter with rated power fault-tolerant feature

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ABSTRACT

Multilevel inverters with fault-tolerance capabilities are critical for powering modern emergency loads, where reliability is the crucial parameter. For enhanced reliability, this paper introduces a single-phase five-level fault-tolerant multilevel inverter to ensure continuous operation even after the occurrence of the faults, while maintaining rated power output. The proposed converter achieves this feature by adopting redundant switches and triacs, which improves the fault-tolerant capability. This paper presents the detailed operating principle of the proposed converter, and its effectiveness is validated through the MATLAB/SIMULINK platform. Further, a 500 W prototype is developed and tested under normal and fault conditions. The experimental results confirmed that the proposed converter can be reconfigured to operate at rated power during faulty conditions. Furthermore, an extensive analysis of reliability, efficiency, and other performance metrics is presented to evaluate the superiority of the proposed converter. The advantages of the proposed converter make it an excellent choice for critical applications.

1. Introduction

Multilevel inverters (MLIs) possess numerous merits such as reduced voltage stress, lower harmonic distortion, and improved electromagnetic compatibility, making them suitable for applications in AC motor drives, FACTS, electric vehicles, and renewable energy integration [1–4]. Some of the specific companies that use multilevel inverters (MLIs) are ABB Group for industrial drives and power systems [5], Siemens for renewable energy and industrial automation [6], General Electric for wind power and grid solutions [7], Schneider Electric for industrial automation and power management [8], Mitsubishi Electric for HVAC and power systems [9], etc. These companies are leaders in the industries that utilize features of MLIs for efficient energy conversion in medium- to high-voltage applications. The cascaded half-bridge (HB) structure is a well-known topology that comes with its modular design and redundant switching states [10]. But with an increase in the number of voltage levels the required higher number of switches and isolated voltage sources are the major challenges [11]. Other well-known structures in the inverter family are neutral-point-clamped [12], Flying capacitor [13], and T-type structures [14]. These structures can offer a higher number of voltage levels with a single DC source; they contain, a higher number of semiconductor switches, and DC-link capacitors or flying capacitors to split the DC-source for generating the higher voltage levels. Here, maintaining the capacitor's voltages balanced and the associated complexity is a major challenge in the MLIs.

Further, with the development and advancement in the converters, the research is directed toward reducing the number of semiconductor devices and enhancing the converter's efficiency. However, the reduced device count in multilevel inverters (MLIs) can compromise modularity and the symmetrical voltage stress distribution on switches. It may result in the loss of redundant switching states, which are essential for fault-tolerant operation. A survey of reduced device count MLIs has been presented in [15,16] highlighting the converter's advantages, limitations, and device counts. On the other hand, the converters in [17]-[20] contain few switches that experience more than two times higher voltage stress than other switches.

In recent decades, researchers developed switched capacitor-based MLIs, which help in generating boosted output [21]-[23]. In [21] by utilizing 8 switches, three diodes, two switched capacitors and one DC source, a 9-level MLI is presented. Here, the converter gives two times

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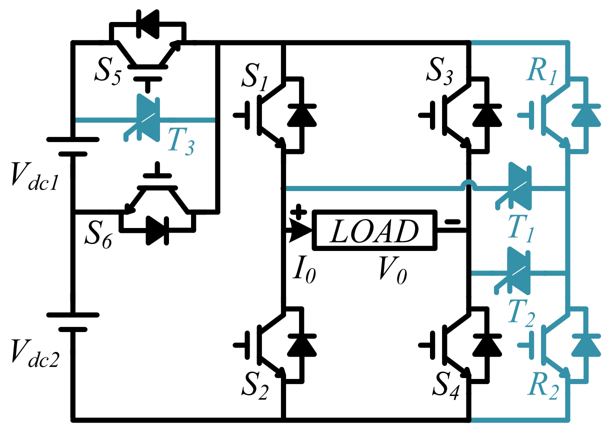


Fig. 1. Proposed Five level FT MLI.

boosting and also can be extended to higher voltage levels by utilizing the level-generator cells. A similar work is presented in [22], by utilizing the lesser number of switches but generating a three times boosting of the input voltage. This structure also uses the voltage level generator and HB circuit for the polarity reversal. The converter in [21] and [22] has the same problem of higher voltage stress of the HB switches. A modular MLI is proposed in [23], which does not utilize an HB circuit, and by using one DC source a flying capacitor the converter generates a five-level output voltage. Further, the converter has inherent capacitor voltage balancing. In an inverter system, the semiconductor devices are more fragile in nature [24]; moreover, few switches in [15–23], conduct for either of the half cycles which means that failure of that switch may lead to failure of the total converter system. Furthermore, considering applications like remote military systems, aircraft, offshore wind energy, and ship propulsion, require highly reliable inverters to provide a continuous power supply even during faults. Consequently, deploying these converters in such applications is not advisable. Therefore, current research is focused on developing MLI topologies that incorporate fault-tolerant features while keeping switch counts minimum.

In a converter system, fault tolerance can be achieved through redundant states and reconfiguration of the control strategy, with some approaches involving modifications to conventional topologies or the introduction of new fault-tolerant designs [25,26]. Based on partial fault tolerance, a few converters have been presented in [27]-[29], which do not use any additional redundant switches but provide fault tolerance to a few fault cases. In [27], by employing one NPC and one T-type structure a five-level multilevel inverter is proposed. This converter can provide single-switch fault tolerance during the post-fault but with half-rated power. However, multi-switch fault tolerance is not possible here and power loss is higher due to the higher number of switching components. With a lesser number of switches fault-tolerant MLIs are proposed in [28] and [29]. In [28], two T-type structures are utilized with two DC sources, which feature additional redundant switching paths for middle voltage levels. Here, with the occurrence of switch faults, the redundant paths are used to reconfigure the converter during the post-fault. Nevertheless, the fault tolerance is limited and rated power output is not possible in most of the fault cases. Similar to the above work, [29] presents a modular MLI that offers fault tolerance through redundant switching paths. But this configuration has the demerits of the requirement of higher isolated DC sources. Additionally, some switching states connect DC sources in reverse series, which is impractical. In converters [30–33], nine-level reduced switch counts MLIs have been presented which works in both symmetrical and asymmetrical voltage levels and uses four DC sources. Here, [30] and [31] contain developed HB circuits which result in higher total standing voltage (TSV). Considering improvement to [30] and [31], f-type and NPC-type-based MLIs are proposed in [32] and [33] respectively. Here, [32] uses the same number of switches but the TSV of the converter decreases. Similarly, [33] has reduced TSV but contains the same number of switches. However, it comes with an extra 4 clamping diodes and increased concerns about reliability. Further, these converters [30–33] have a common issue of asymmetric voltage stress where two switches have 4 times the voltage stress than other switches. Furthermore, the conduction time of highly stressed switches belongs to either of the power cycles. Consequently, parallel redundant switches are used across those switches, increasing the fault tolerance. However, the parallel connection of the redundant switches should be avoided to avoid the same voltage stress. Another issue with These converters is, they cannot provide the rated power in each fault case.

To improve the reliability of the converters and to provide the rated power during the post-fault a few works have been presented [34–41]. In [34], a flying capacitor-based five-level fault tolerant inverter is presented which uses two redundant units and each unit contains a half-bridge. Here, the converter features preserved rated power fault tolerance. A similar work that uses the same number of main and redundant switches is presented in [35]. Here, two DC sources are utilized to generate a five-level voltage output. However, a common issue in [34] and [35] is the use of redundant switches connected directly parallel to the main switches. In [36–38], HB-type, F-type, and NPC-type redundant legs are utilized to introduce fault tolerance. Here, fault tolerance is possible for all the switch fault cases, but these converters use a higher number of switches. Further, the parallel connection of the redundant switches in [36–38] is another issue. By utilizing a t-type leg and an NPC-type leg a five-level MLI is proposed in [39], and to make the converter resilient, an additional T-type is utilized which is connected to both the t-type leg and NPC-type main legs. Here the converter has single and multi-switch fault resilience but at the cost of higher TSV and higher switches. To decrease the TSV of [39], the T-type redundant unit is replaced with an Active NPC-type leg in [40]. However, it increases further the converter's switches and hence the power losses. Similarly, NPC-type legs are used in [41] to generate a five-level output and use an extra half-bridge and an NPC-type leg as the redundant legs. Here, the redundant unit is helpful in addressing the faults with rated power but at the cost of the higher number of switches and TSV.

To address the aforementioned issues, this work proposes a novel single-phase fault-tolerant converter, it comprises six main switches operated to generate a five-level voltage output. In addition, the proposed converter uses two redundant MOSFET/IGBT switches and three TRIACs, which provide additional switching paths to enhance fault tolerance. The TRIACs in the converter operate after faults occur and have a lower probability of failure compared to the MOSFETs/IGBTs. A key feature of the proposed converter is its use of fewer main and redundant switches, while still offering tolerance to single and multiple switch faults at rated power.

2. Proposed five-level FT MLI

2.1. Proposed FT MLI configuration

The configuration of the proposed converter is shown in Fig. 1. This converter includes six main switches and five redundant switches. All six main switches (S_1, S_2, S_3, S_4, S_5, and S_6) are MOSFETs/IGBTs, and among the five redundant switches, two are MOSFETs/IGBTs (R_1, and R_2), while the remaining three are TRIACs (T_1, T_2, and T_3). As shown in Fig. 1, the direct parallel connection of the redundant switches to the main switches is avoided by using TRIACs. This design minimizes the voltage stress on the redundant switches during normal operation and ensures that they remain unaffected when the main switches encounter faults. The main switches operate during healthy conditions, whereas the redundant switches are utilized during fault conditions. When the converter is operated with DC supplies $V_{dc1} = V_{dc2} = V_{dc}$, the voltage stress on both the main and redundant switches under healthy

Table 1
Switching States Consisting of Main Switches.

Main switch states	S_1	S_2	S_3	S_4	S_5	S_6	V_{LOAD}
M_1	1	0	0	1	1	0	+2V
M_2	1	0	0	1	0	1	+V
M_3	0	1	0	1	0	0	0 V
M_4	1	0	1	0	0	0	0 V
M_5	0	1	1	0	0	1	−V
M_6	0	1	1	0	1	0	−2V

conditions can be described as follows:

$$V_{S1} = V_{S2} = V_{S3} = V_{S4} = 2V_{dc} \tag{1}$$

$$V_{S5} = V_{S6} = V_{R1} = V_{R2} = V_{T1} = V_{T2} = V_{T3} = V_{dc} \tag{2}$$

The voltage stress on the main switches remains the same in the post-fault reconfiguration but the voltage stress of a few redundant switches changes which be described as follows:

$$V_{R1} = V_{R2} = V_{T1} = V_{T2} = 2V_{dc} \tag{3}$$

2.2. *Healthy mode of operation*

During healthy operation, the proposed converter generates a five-level output voltage (+2V, +V, 0, −V, and −2V), using the main switch states provided in Table 1. The circuit diagram illustrating each voltage level is shown in Fig. 2. From this figure, the converter's healthy operation can be categorized into six modes or switching states:

Mode-1 (+2V): As depicted in Fig. 2(a), switches S_1, S_4, and S_5 form the conduction path to generate the + 2 V voltage level. Three switches are involved in the conduction path to achieve this voltage.

Mode-2 (+V): In this mode, switches S_1, S_4, and S_6 create the conduction path, generating voltage level + V, as shown in Fig. 2(b). Similar to Mode 1, three switches are used to generate this voltage level.

Mode-3 (0): As indicated in Table 1 and illustrated in Fig. 2(c), switches S_2 and S_4 participate in the conduction path to generate a 0 V output. Only two switches are required in this case.

Mode-4 (0): This mode also generates a 0 V output, involving two switches in the conduction path. However, in this case, switches S_1 and S_3 are engaged in the conduction path.

Mode-5 (−V): As shown in Fig. 2(e), switches S_2, S_3, and S_6 conduct to generate a −V voltage level. Here, negative current flows through the load.

Mode-6 (−2V): According to Table 1, the switch configuration for this Mode involves S_2, S_3, and S_5 in the conduction path to generate a −2V voltage level. The circuit diagram for generation_2V voltage level is given in Fig. 2(f).

In this work, fixed reference frequency ($f_{reference}$), carrier frequency ($f_{carrier}$), and modulation index (M_a) level shift pulse width modulation (PWM) techniques have been implemented as depicted in Fig. 3.

The proposed work can also be implemented in a three-phase system as shown in Fig. 4.

2.3. *Fault operation*

When faults occur in the main switches, some voltage levels are

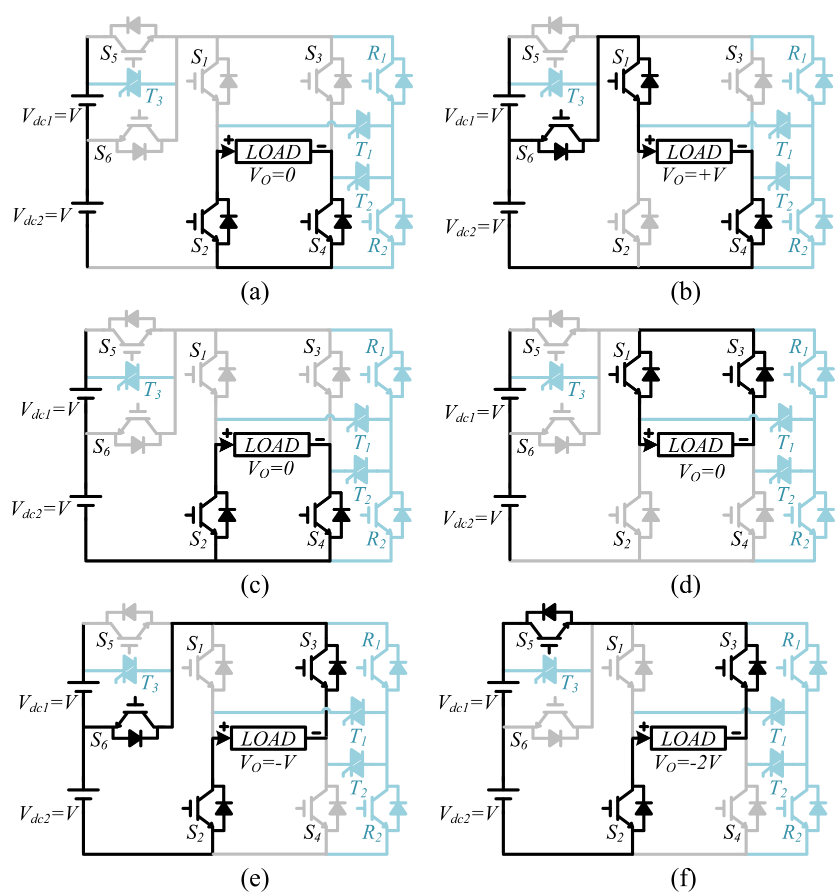


Fig. 2. Circuit diagram for voltage level (a) + 2 V(M_1), (b) + V(M_2), (c) 0 V (M_3), (d) 0 V (M_4), (e) −V(M_5), (f) −2V(M_6).

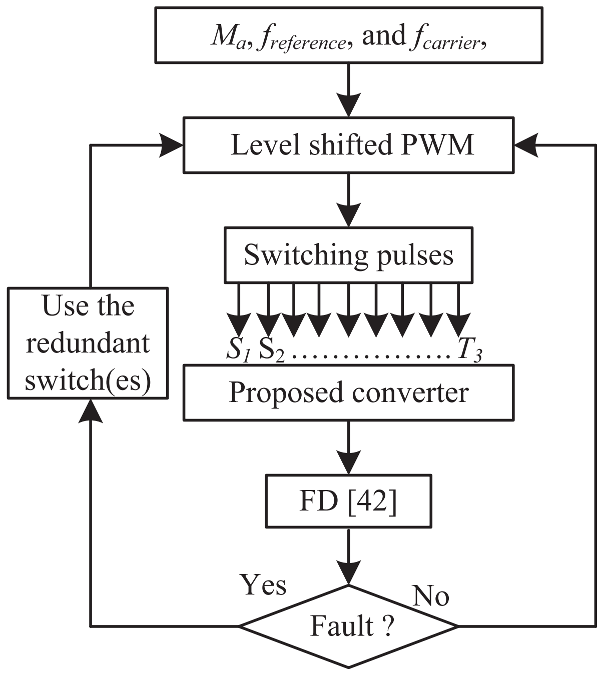


Fig. 3. Flow chart of the control algorithm.

affected, as indicated in Table 1. Once a fault signal is detected by the method outlined in [42], the redundant switches can be used to reconfigure the converter for post-fault operation, as depicted in Fig. 2 and given in Table 2. As given in Table 1, when switch S_1 becomes faulty, the switching States M_1 for voltage level + 2 V and M_2 for voltage level + V become unavailable; in this scenario, the switches S_5, R_1, T_1, and S_4 are used to generate voltage level + 2 V, and S_6, R_1, T_1, and S_4 are used to get the voltage level + V. Similarly, with faults on switches S_1 and S_2, all the voltage levels are affected. Here, the unavailable can be made available using the redundant switching paths given in Table 2; for the + 2 V voltage level, switches S_5, R_1, T_1, and S_4 can be utilized; for the + V level, switches S_6, R_1, T_1, and S_4; for the 0 V level, switches R_1, T_1, and S_3; for the −V level, switches S_6, S_3, T_1, and R_2; and for the −2V level, switches S_5, S_3, T_1, and R_2 can be used. For better understanding, TRIAC T_1 is deployed in the event of a fault within the half-bridge leg comprising switches S_1 and S_2. Likewise, TRIAC T_2 is engaged when a fault occurs in the half-bridge leg containing switches S_3 and S_4, while TRIAC T_3 is utilized in case of a fault on switch S_5.

As illustrated in Table 2, the switching paths remain available to generate the rated power for all combinations of single and dual switch faults, except in four specific cases (S_1S_3, S_1S_4, S_2S_3, and S_2S_4).

It is noted that this work primarily focuses on the fault-tolerant strategy for open circuit (OC) faults, with fault signals emulated by applying a zero-gate signal to the switch considered faulty.

3. Simulation and hardware results discussion

The fault tolerance feature of the proposed converter is validated using MATLAB-SIMULINK software and a physical prototype. A 500 W prototype with rating given in Table 3 is fabricated for hardware testing as shown in Fig. 5. For operating the proposed converter, the switches are controlled by the FPGA controller with the LSPWM technique, operating under both normal and OC fault conditions. Here, one healthy case, four single-switch faulty cases, and four double-switch fault cases have been considered and discussed below.

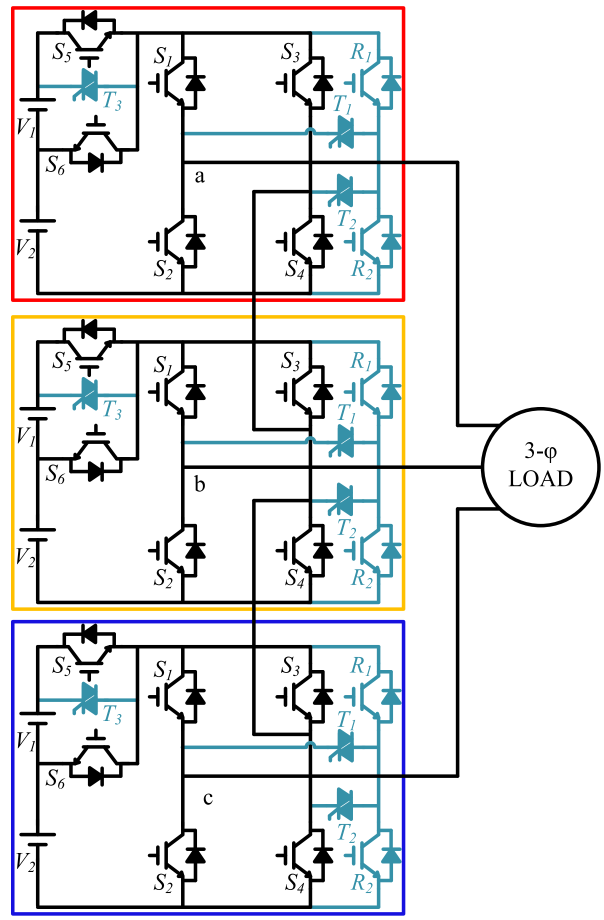


Fig. 4. Three-phase implementation of the proposed converter.

Table 2
Switching States with Redundant Switches.

Faulty Switches	ON states swathes for different voltage levels				
	+2V	+V	0	−V	−2V
S_1	$S_5R_1T_1S_4$	$S_6R_1T_1S_4$	S_2S_4	$S_6S_3S_2$	$S_5S_3S_2$
S_2	$S_5S_1S_4$	$S_6S_1S_4$	S_1S_3	$S_6S_3T_1R_2$	$S_5S_3T_1R_2$
S_3	$S_5S_1S_4$	$S_6S_1S_4$	S_2S_4	$S_6R_1T_2S_2$	$S_5R_1T_2S_2$
S_4	$S_5S_1T_2R_2$	$S_6S_1T_2R_2$	S_1S_3	$S_6S_3S_2$	$S_5S_3S_2$
S_5	$T_3S_1S_4$	––	S_1S_3/S_2S_4	––	$T_3S_3S_2$
S_6	$S_5S_1S_4$	––	S_1S_3/S_2S_4	––	$S_5S_3S_2$
S_1S_2	$S_5R_1T_1S_4$	$S_6R_1T_1S_4$	$R_1T_1S_3/$ $T_1R_2S_4$	$S_6S_3T_1R_2$	$S_5S_3T_1R_2$
S_1S_3	––	––	––	––	––
S_1S_4	––	––	––	––	––
S_1S_5	$T_3R_1T_1S_4$	––	S_2S_4	––	$T_3S_3S_2$
S_1S_6	$S_5R_1T_1S_4$	––	S_2S_4	––	$S_5S_3S_2$
S_2S_3	––	––	––	––	––
S_2S_4	––	––	––	––	––
S_2S_5	$T_3S_1S_4$	––	S_1S_3	––	$T_3S_3T_1R_2$
S_2S_6	$S_5S_1S_4$	––	S_1S_3	––	$S_5S_3T_1R_2$
S_3S_4	$S_5S_1T_2R_2$	$S_6S_1T_2R_2$	S_2S_4	$S_6R_1T_2S_2$	$S_5R_1T_2S_2$
S_3S_5	$T_3S_1S_4$	––	S_2S_4	––	$T_3R_1T_2S_2$
S_3S_6	$S_5S_1S_4$	––	S_2S_4	––	$S_5R_1T_2S_2$
S_4S_5	$T_3S_1T_2R_2$	––	S_1S_3	––	$T_3S_3S_2$
S_4S_6	$S_5S_1T_2R_2$	––	S_1S_3	––	$S_5S_3S_2$
S_5S_6	$T_3S_1S_4$	––	S_1S_3/S_2S_4	––	$T_3S_3S_2$

Table 3
Experimental Setup Parameters.

Parameters	
DC sources ($V_{dc1} = V_{dc2}$)	50 V
Reference frequency ($f_{reference}$)	50 Hz
IGBT KGF25N120	1200 V, 25 A
TRIAC BTA20-600CWRG	600 V, 10 A
Load Resistance, Inductance	10 Ω, 10 mH
Gate driver IC	TLP250
Controller	FPGA SPARTAN-6
Carrier frequency ($f_{carrier}$)	10 kHz

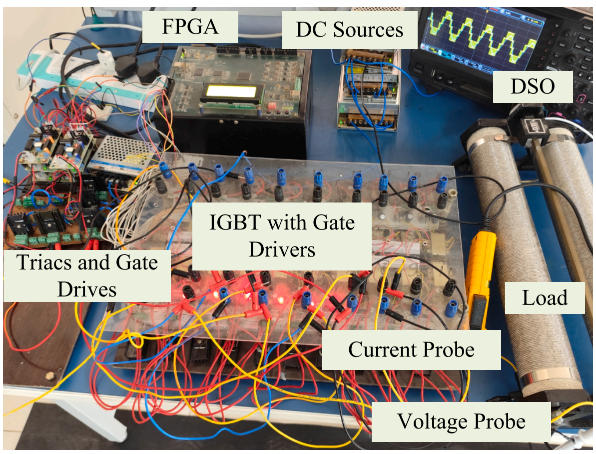


Fig. 5. Photograph of the experimental prototype.

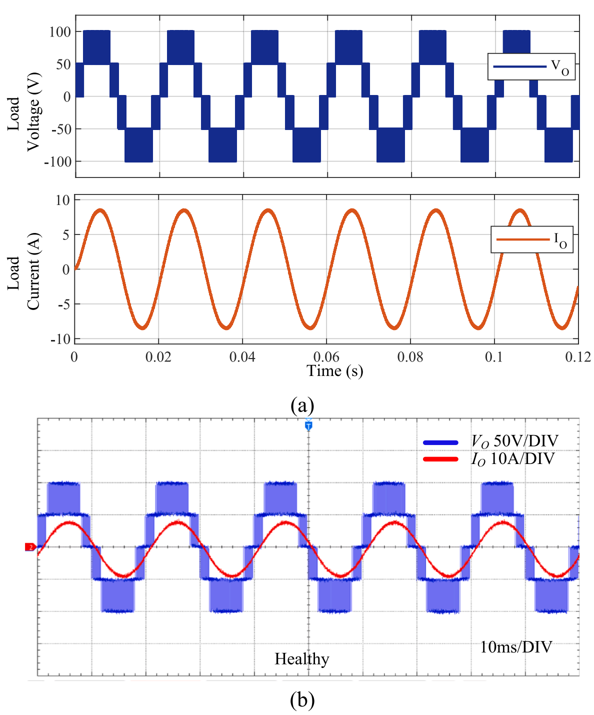


Fig. 6. Load voltage and load current under healthy conditions (a) simulation result (b) hardware result.

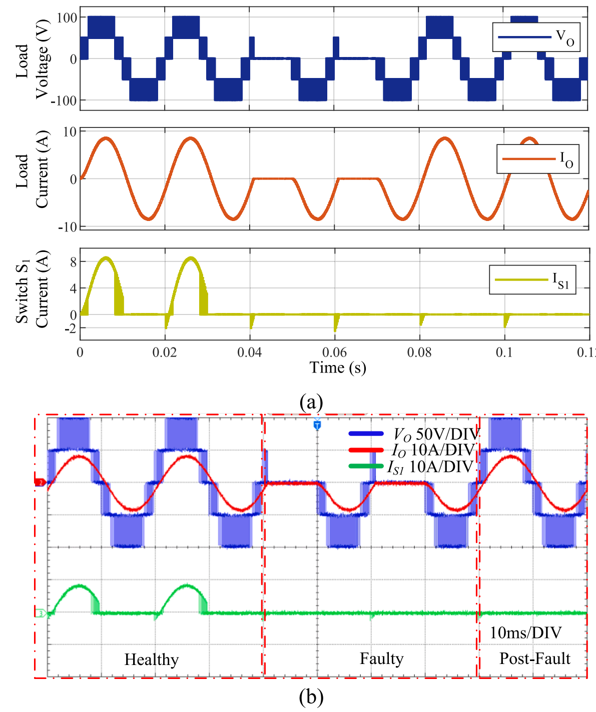


Fig. 7. Load voltage, load current and switch current under fault on switch S_1 (a) simulation result (b) hardware result.

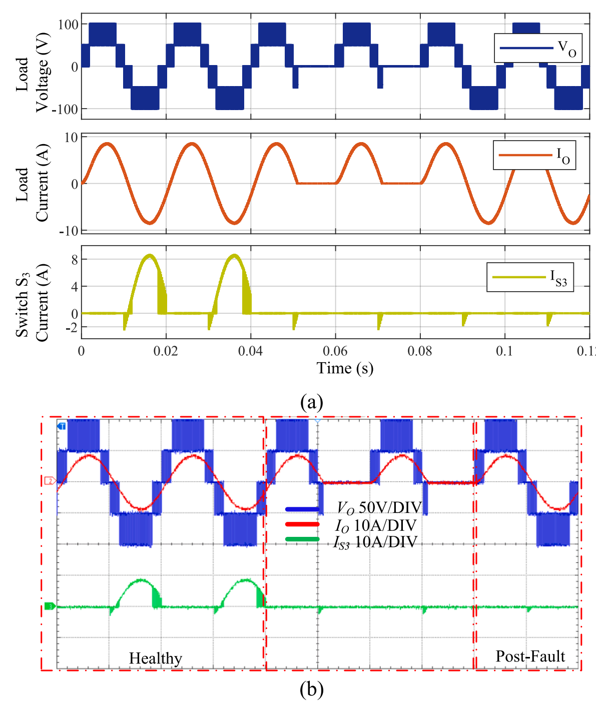
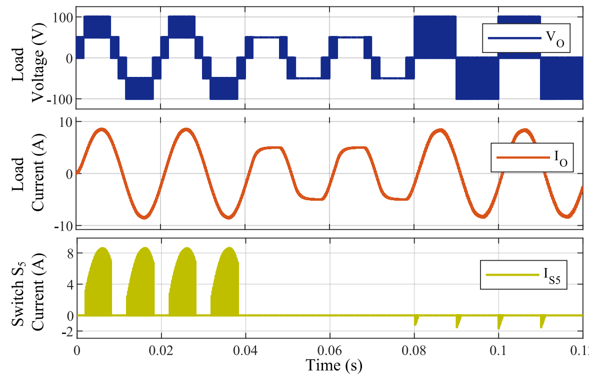
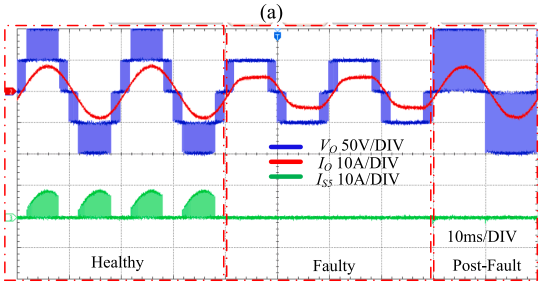


Fig. 8. Load voltage, load current and switch current under fault on switch S_3 (a) simulation result (b) hardware result.

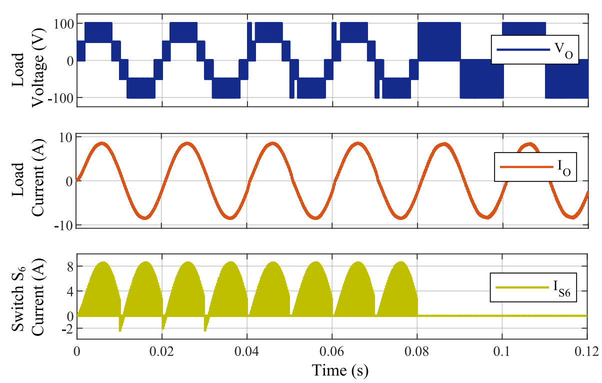


(a)

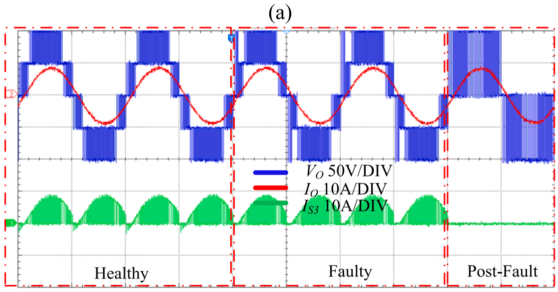


(b)

Fig. 9. Load voltage, load current and switch current under fault on switch S_5 (a) simulation result (b) hardware result.

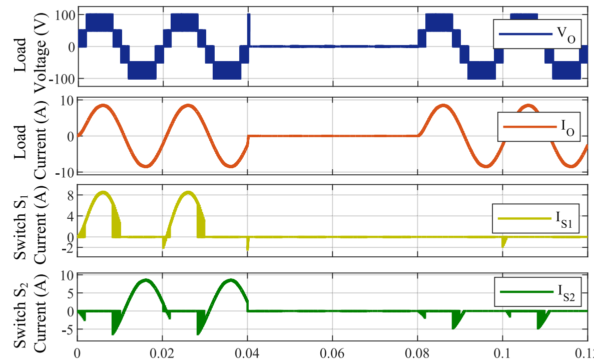


(a)

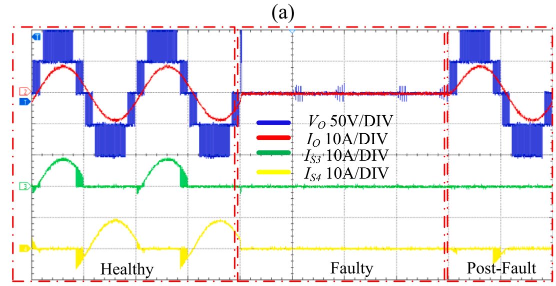


(b)

Fig. 10. Load voltage, load current and switch current under fault on switch S_6 (a) simulation result (b) hardware result.



(a)



(b)

Fig. 11. Load voltage, load current and switch current under fault on switch S_1 S_2 (a) simulation result (b) hardware result.

3.1. Healthy operation

When the proposed converter operates under healthy conditions with the switching states given in Table 1, it generates a 5-level voltage. The simulation and hardware results showcasing the output voltage and current are shown in Fig. 6(a) and (b), respectively.

3.2. Fault operation

3.2.1. S_1 OC Fault

As shown in Fig. 7(a) and (b), the switching paths for voltage levels + 2 V and + V are eliminated when a fault on switch S_1 is emulated. Once the fault is identified within two power cycles (0.04 s), the redundant switches R_1 and T_1 are employed to generate the previously unavailable + 2 V and + V voltage levels, as indicated in Table 2. The switching states for other voltage levels remain unchanged. By utilizing the switching states provided in Table 2, the LSPWM is reconfigured, and the converter produces a five-level output, as illustrated in Fig. 7(a) and (b).

3.2.2. S_3 OC Fault

Similar to the OC fault on S_3, when a fault occurs on switch S_3 the negative voltage levels are eliminated as shown in Fig. 8(a) and (b). After the fault detection, the redundant switches R_2 and T_1 are used to reconfigure the negative voltage levels, as indicated in Table 2. However, the switching states for other voltage levels remain unchanged. Here, after the post-fault reconfiguration the converter produces a five-level output, as illustrated in Fig. 8(a) and Fig. 8(b).

3.2.3. S_5 OC Fault

Switch S_5 is involved in the switching paths for generating the peak voltage levels ± 2 V. Therefore, when a fault is introduced in switch S_5 by setting its gate signal to zero, the proposed converter is unable to produce the ± 2 V voltage levels, as shown in Table 1. Upon detection of the fault, the TRIAC T_3 is activated at the reconfiguration instant to

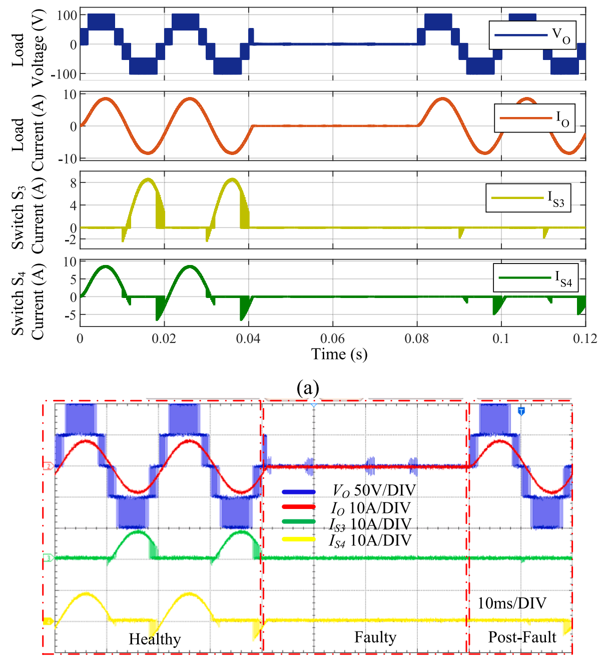


Fig. 12. Load voltage, load current and switch current under fault on switch S_3 S_4 (a) simulation result (b) hardware result.

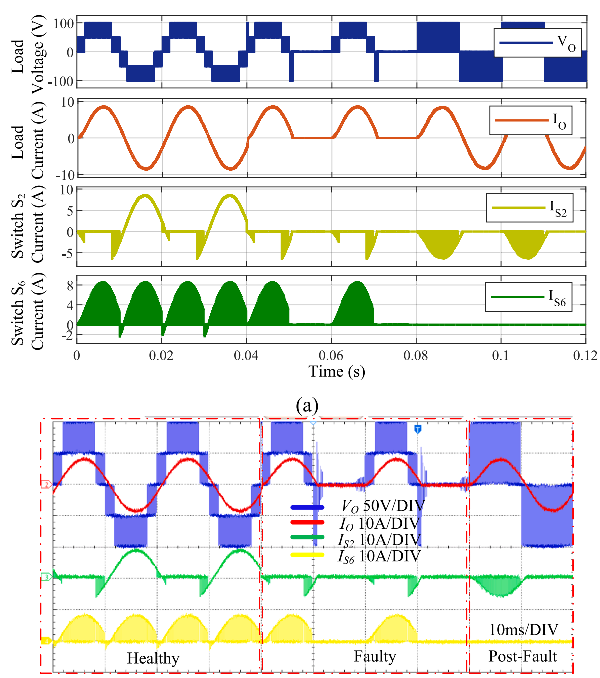


Fig. 13. Load voltage, load current and switch current under fault on switch S_2 S_6 (a) simulation result (b) hardware result.

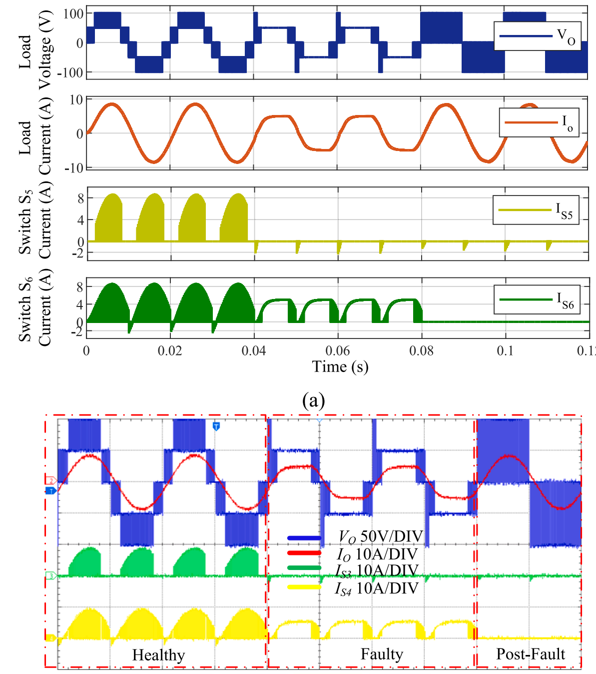


Fig. 14. Load voltage, load current and switch current under fault on switch S_5 S_6 (a) simulation result (b) hardware result.

restore the missing voltage levels. However, when TRIAC T_3 is activated, switch S_6 cannot be turned on without causing a short circuit of the source V_{dc1}. Consequently, generating the voltage levels ± V is not feasible. Despite this, the converter reconfiguration with voltage levels ± 2 V and 0 still allows it to deliver the rated power. The output simulation and experimental results for this fault scenario are presented in Fig. 9(a) and Fig. 9(b).

3.2.4. *S_6 OC Fault*

This fault case is very similar to the previous fault case i.e., fault on Switch S_5. Here, as shown in simulation and hardware results depicted in Fig. 10(a) and (b), the switch fault should result in loss of voltage levels ± V; however, due to the S_6 body diode, these voltage levels remain available but has little impacts on the voltage and waveforms. From the figures, the current waveform gets slightly distorted. In this case, during the post-fault, TRIAC T_3 is continuously turned on and the converter is reconfigured with the reduced voltage levels ± 2 V and 0. However, rated power and better % THD are ensured as depicted in simulation and experimental results.

3.2.5. *S_1S_2 OC Fault*

As depicted in Fig. 1, switches S_1 and S_2 are part of the same leg and operate in a complementary manner. The redundant leg switches R_1 and R_2 are connected parallel with the main switches leg; this configuration allows the converter to provide redundant paths for both switches simultaneously. Therefore, when faults occur in switches S_3 and S_4 simultaneously, all voltage levels are affected, as shown in Fig. 11(a) and (b). During the post-fault reconfiguration, redundant switches R_1 and R_2 take over the functions of switches S_1 and S_2, thereby restoring the same voltage levels and rated power output. Additionally, TRIAC T_1 is activated in this reconfiguration process.

3.2.6. *S_3S_4 OC Fault*

Similar to the previous case, switches S_3 and S_4 are part of the same

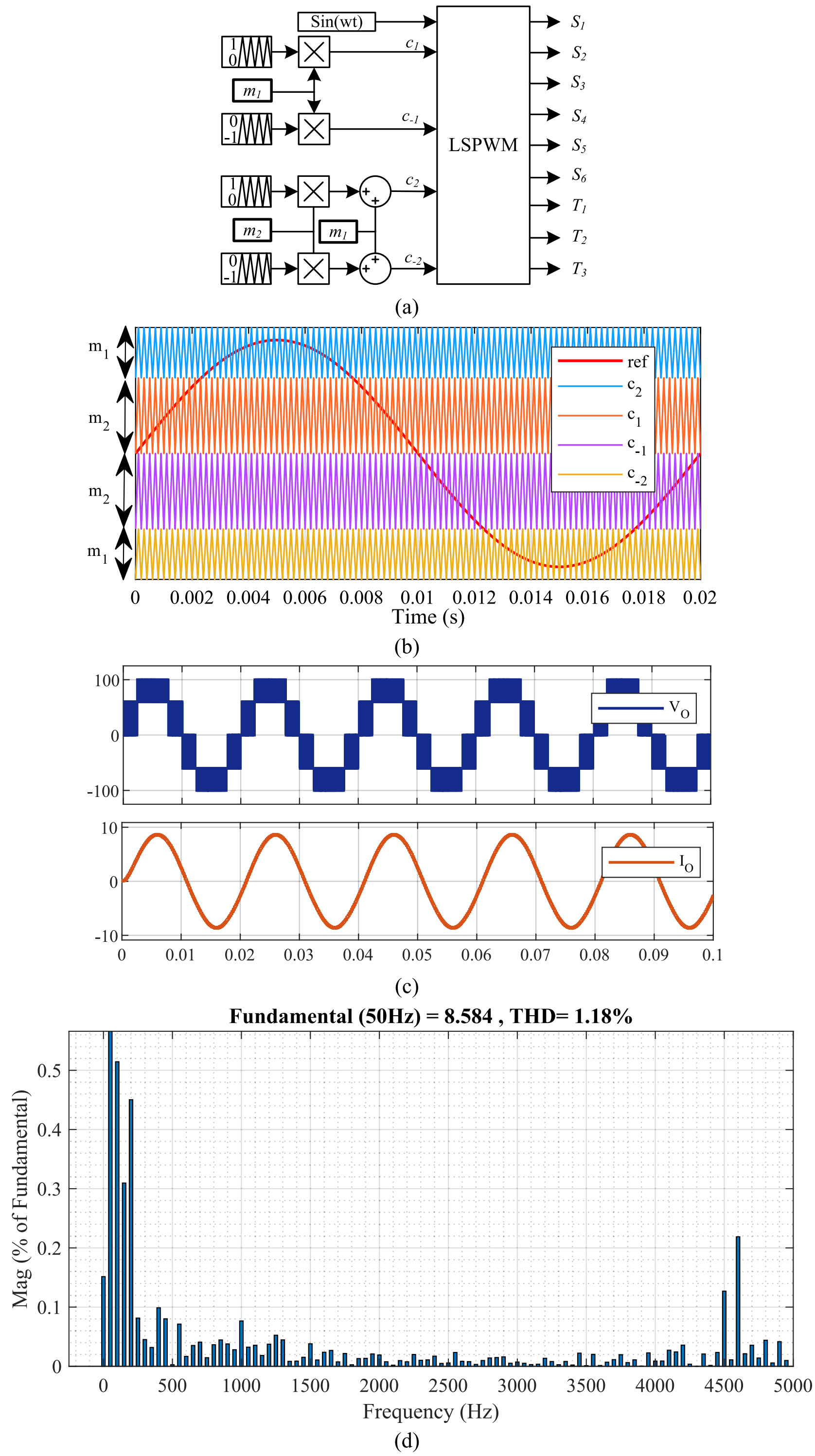


Fig. 15. (a) Modified LSPWM under DC source voltage fluctuation (b) modified carrier and reference signals (c) simulated load voltage and load current (d) % THD.

Table 4
Power Loss and Efficiency for Different Modulation Index.

Modulation Index	Total power loss(W)	Efficiency (%)
0.1	1.14	89.76
0.2	2.49	94.13
0.3	3.52	96.23
0.4	4.55	97.23
0.5	5.58	97.81
0.6	6.66	98.18
0.7	8.84	98.22
0.8	10.05	98.45
0.9	11.29	98.62
1	12.65	98.75

Table 5
Power Loss and Efficiency for Different Output Power.

Output power (W)	Total power loss(W)	Efficiency (%)
100	1.7	98.32
200	2.77	98.63
400	4.87	98.79
600	6.93	98.85
800	9.75	98.79
1000	12.65	98.75
1200	15.27	98.74
1400	18.18	98.71
1600	21.24	98.68
1800	24.49	98.65
2000	27.91	98.62

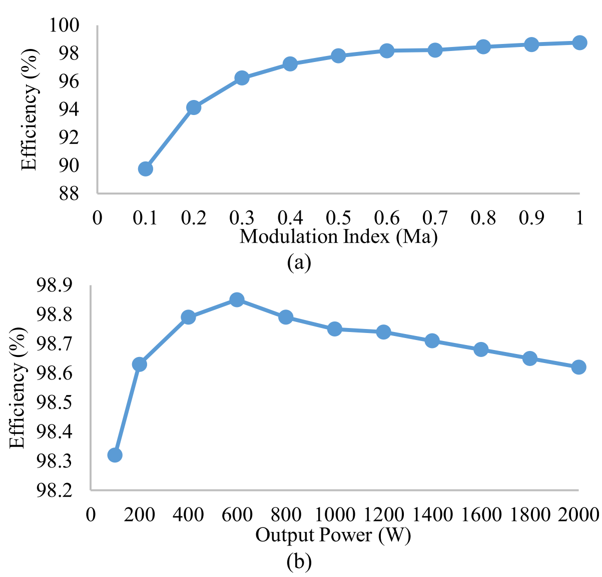


Fig. 16. Healthy Efficiency curve against (a) modulation index and (b) output power.

leg which operates in a complementary manner and contributes to all the voltage levels. Here, both switches of the redundant leg can be simultaneously accessed by activating the TRIAC *2* when S_3 and S_4 get simultaneous faults. Therefore, when faults occur in switches S_3 and S_4, all voltage levels are affected, as shown in Fig. 12(a) and (b). During the post-fault reconfiguration, redundant switches R_1 and R_2 take over the functions of switches S_3 and S_4, thereby restoring the same voltage levels and rated power output. Here, TRIAC T_2 is activated in the reconfiguration process.

3.2.7. S_2S_6 OC Fault

As indicated in Table 1, if switching state M_3 is not used during normal operation, a fault in switches S_2 and S_6 will eliminate the voltage levels $\pm$ V and $-$2V. The converter lacks a redundant path for switch S_6 but has a redundant path for switch S_2. Therefore, upon the fault detection, the control strategy outlined in Fig. 2 is implemented: the voltage levels $\pm$ V are omitted, and the level-shift PWM is changed by replacing the control signal from the faulty S_2 to the redundant switch R_2, also TRIAC T_1 is activated. The results corresponding to faults in switches S_2 and S_6 are depicted in Fig. 13(a) and (b). This reconfiguration allows the proposed converter to produce a three-level rated power output.

3.2.8. S_5S_6 OC Fault

Switches S_5 and S_6 serve as voltage selectors, i.e., switch S_5 is turned ON to access the $\pm$ V voltage levels, and switch S_6 is turned ON to get the $\pm$ 2 V voltage level. So, when the OC switch faults occur on switches S_5 and S_6 the voltage levels $\pm$ V and $\pm$ 2 V become available as per Table I. However, as depicted in Fig. 14(a) and (b), the voltage levels $\pm$ V are seen to be available due to forward biasing of the body diode of switch S_6. In this fault case, after the detection of the fault, the control strategy outlined in Fig. 2 is implemented. Wherein, the voltage levels $\pm$ V are discarded, and the level-shift PWM reconfigured to generate the rated power with three voltage levels 0 and $\pm$ 2 V. Here, TRIAC T_3 is continuously turned ON to recreate the path broken by switch S_5: the corresponding results are depicted in Fig. 14(a) and (b).

From the above discussion of 8 faulty cases, it is clear that the proposed converter is able to handle all switch faults while generating the rated power during the post-fault. However, from the discussion, it noted that few fault case restoration is possible with rated power but reconfigured with a reduced number of voltage levels.

4. Converter's operation under fluctuation in independent DC input source voltages

When the DC input source voltages of the proposed converter fluctuate, the converter's THD is affected if the general LSPWM technique is employed. However, the effect on the THD can be minimized by modifying the LSWPM as given in Fig. 15(a). From the figure, the carrier's height is modified with fluctuation in the DC source voltages. Here, if V_{dc1} and V_{dc2} are defined as the DC source voltages, the carrier's height shown in Fig Y, can be given as:

$$m_1 = \frac{V_{dc1}}{V_{dc1} + V_{dc2}} \tag{4}$$

$$m_2 = \frac{V_{dc1}}{V_{dc1} + V_{dc2}} \tag{5}$$

By considering a scenario when the independent DC input source voltages fluctuate to $V_{dc1} = 60$ V and $V_{dc2} = 40$ V. In this case, m_1 and m_2 can be given as 0.4 and 0.6, respectively. Now, when the proposed converter is configured with the LSPWM with c_1, c_2, c_{-1}, and c_{-2} carriers shown in Fig. 15(b), the converter generates the output as depicted in Fig. 15(c). From the figure, the load voltage has five voltage levels 100, 60, 0, $-$60, and $-$100 and the current remains nearly sinusoidal. Further, the current THD profile is plotted for this case as shown in Fig. 15(d). From the figure, it is clear that the current THD remains minimal similar to when the converter operates with equal DC source voltages.

It is important to mention that the suggested method needs the information of independent DC input source voltages which require two voltage sensors at the input DC sources.

5. Power loss, efficiency, and THD analysis

The power loss of a power converter depends on the switching and

Table 6
Power Loss, Efficiency, and THD for Healthy and Different Fault Cases.

Converter's state	$P_{CN_{LOSS_{total}}}$ (W)	$P_{SW\_LOSS\_total}$ (W)	$P_{LOSS_{total}}$ (W)	% Efficiency	% Voltage THD	% Current THD
Healthy	12.29	0.36	12.65	98.75	2.1	36.53
S_1_fault	13.29	0.36	13.65	98.65	2.1	36.53
S_3_fault	13.29	0.36	13.66	98.65	2.1	36.53
S_5_fault	16.48	0.99	17.47	98.28	2.2	66.46
S_6_fault	16.48	0.99	17.47	98.28	2.2	66.46
S_1S_2_fault	13.29	0.36	13.65	98.65	2.1	36.53
S_3S_4_fault	13.29	0.36	13.65	98.65	2.1	36.53
S_2S_6_fault	17.64	0.99	18.63	98.17	2.2	66.46
S_5S_6_fault	16.48	0.99	17.47	98.28	2.2	66.46

Fig. 17. For healthy and different fault conditions (a) conduction, switching, and total power loss and (b) efficiency, and (c) % voltage and current THD.

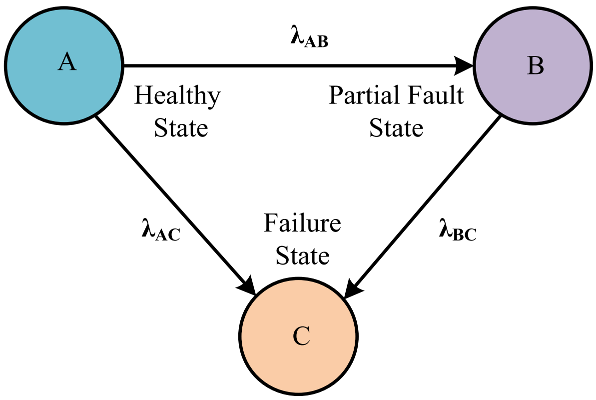


Fig. 18. Markov chain model with three transition states.

Table 7
Failure Rate of Switches.

Failure rate of Semiconductor Devices	Failure rate per 10^6 Hours
λ_{IGBT}	2.13
λ_{TRIAC}	0.911

Table 8
Transition Failure Rate.

Proposed Converter	Transition Failure rates
Without redundant switches	$\lambda_{AB} = \lambda_S$
	$\lambda_{BC} = 5\lambda_S$
	$\lambda_{AC} = 5\lambda_S + 15\lambda_S^2$
	$\lambda_{AA} = 6\lambda_S + 15\lambda_S^2$
	$\lambda_{BB} = 5\lambda_S$
With redundant switches	$\lambda_{AB} = 6\lambda_S + 11\lambda_S^2$
	$\lambda_{BC} = 2\lambda_S + \lambda_T$
	$\lambda_{AC} = 4\lambda_S^2$
	$\lambda_{AA} = 6\lambda_S + 15\lambda_S^2$
	$\lambda_{BB} = 2\lambda_S + \lambda_T$

conduction loss of the switching devices. Where switching loss occurs due to overlapping of voltage and current of a switch during ON to OFF state and OFF to ON state. If the ON state of the period switch and OFF states periods of a switch are indicated as T_{on} and T_{off}, respectively, the switching loss for a switch can be given in (6). Where, V_{off} denotes the voltage that appears across the switch during the OFF state, $f_{carrier}$ is the carrier or switching frequency, I_{on} indicates the turn ON current, I_{off} indicates the current before the switch goes to OFF states, T_{on} is the time period for OFF to ON state, and T_{off} is the time period for the ON to OFF state [15,33].

$$P_{SW\_LOSS} = \frac{1}{6} V_{off} f_{carrier} \left(I_{on} T_{on} + I_{off} T_{off} \right) \tag{6}$$

Where, T_{on} and T_{off} can be obtained from the datasheet of the semiconductor switch.

If a converter has n switches the generalized expression for the switching loss can be given in (7), where, i denotes for ith switch.

$$P_{SW\_LOSS\_total} = \sum_{i=1}^{n} \frac{1}{6} V_{off(i)} f_{carrier} \left(I_{on(i)} T_{on} + I_{off(i)} T_{off} \right) \tag{7}$$

The major contribution to the conduction loss is due to power loss in the on-state parasitic resistance of the switch and the minor contribution is by ON-state voltage and current. Here, if the ON state voltage is considered as V_{on}, the average switch is considered as I_{SW_avg}, and I_{SW_rms} is considered as the rms current through the switch, the conduction loss of the switch can be given in (8). Here, R_{on} is the on-state parasitic resistance.

$$P_{CN\_LOSS} = V_{on} I_{SW_{avg}} + {I_{SW\_rms}}^2 R_{on} \tag{8}$$

Where R_{on} can be obtained from the saturation voltage characteristics given in the datasheet of the semiconductor switch.

The total conduction loss for n switches can be given as:

$$P_{CN\_LOSS\_total} = \sum_{i=1}^{n} \left(V_{on(i)} I_{SW_{avg(i)}} + {I_{SW\_rms(i)}}^2 R_{on} \right) \tag{9}$$

Now using these losses total power loss and efficiency of the converter can be obtained as given in (10) and (11), respectively. where P_{LOSS_total} is the total power loss, P_{output} is the total output power of the converter.

$$P_{LOSS\_total} = P_{SW\_LOSS\_total} + P_{CN\_LOSS\_total} \tag{10}$$

$$Efficiency(\eta) = \frac{P_{output}}{P_{output} + P_{SW\_LOSS\_total} + P_{CN\_LOSS\_total}} \tag{11}$$

In this work, for a 1 kW power rating, PLECS software is used to determine the switching and conduction loss of the converter. Using equations (8) and (9) total power loss and efficiency of the converter are obtained for different modulation indexes (M_a) and are tabulated as given in Table 4 [15,33]. Further, the converter's power losses are obtained under different power ratings and tabulated in Table 5. Using Table 4 and Table 5, the efficiency curve against modulation index and different power are plotted as depicted in Fig. 16(a) and Fig. 16(b). From Fig. 16(a), the converter's efficiency is maximum at peak modulation index and decreases with a decrease in modulation index. From Fig. 16 (b), the proposed converter attains maximum efficiency at a 600 W

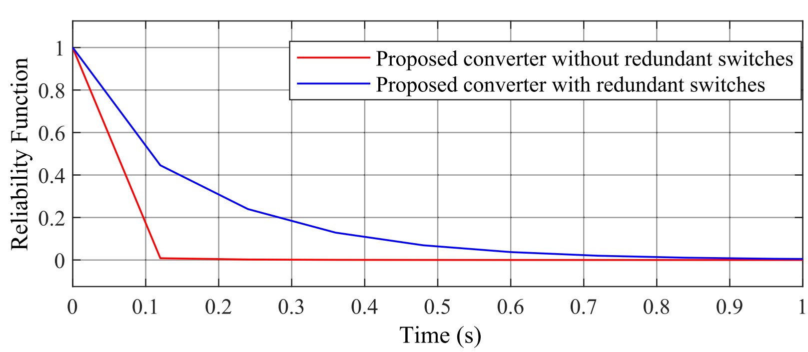


Fig. 19. Reliability curves.

Table 9
Comparison of Recent Five-Level MLIs.

Parameters	[27]	[28]	[29]	[34]	[35]	[36]	[37]	[38]	[39]	[40]	[41]	Proposed
P_1	2	2	4	1	2	1	2	1	1	1	1	2
P_2	7	8	6	8	8	8	8	6	8	8	8	6
P_3	0	0	0	4	4	4	6	6	4	6	6	2
P_4	0	0	0	0	0	0	0	0	0	0	0	3
P_5	6	0	0	0	0	2	4	0	2	2	6	0
P_6	0	0	0	1	0	1	0	1	1	1	2	0
P_7	13	8	6	12	12	14	18	12	14	16	20	11
P_8	15	12	10	18	18	18	22	18	24	22	22	17
P_9	11,500	10,000	9000	17,000	17,000	17,000	19,000	18,000	22,000	20,000	19,000	15,500
P_{10}	97.9	98.4	98.9	97.85	98.35	97.9	97.1	98.2	97.4	97.4	97.1	98.75
P_{11}	X	X	X	✔	✔	✔	✔	✔	✔	✔	✔	✔

P_1 = Dc sources, P_2 = number of main MOSFET/IGBT switches, P_3 = number of redundant MOSFET/IGBT P_4 = number of triacs, P_5 = number of the clamping diode, P_6 = number of DC-link/ flying capacitors, P_7 = a total number of semiconductor devices, P_8 = total standing voltage, P_9 = total switch device power rating (), P_{10} = efficiency, P_{11} = multiple switch fault-tolerant ability.

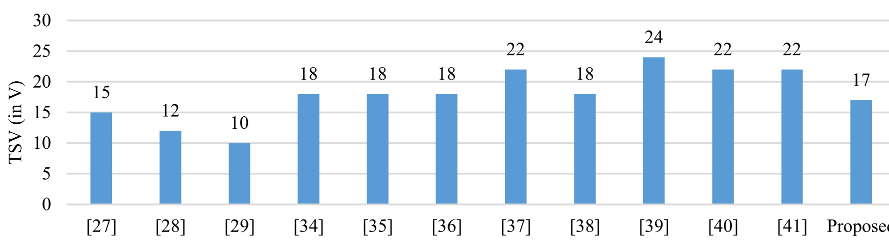


Fig. 20. TSV of the literature and proposed converters.

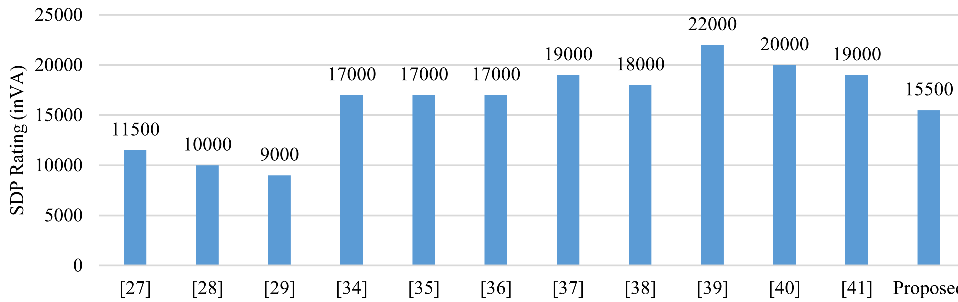


Fig. 21. SDP rating of the literature and proposed converters.

power rating and decreases with an increase/ decrease in the power rating.

Further, to compare the power loss increased due to the inclusion of the redundant switches in the post-fault reconfiguration, the power loss and efficiency analysis of the proposed converter is extended to faulty cases. For healthy and different fault cases, the converter's switching loss, conduction loss, total power loss, and obtained efficiency are tabulated in Table 6. From the table, when the converter is reconfigured after the fault, the conduction losses increase slightly, and switching loss remains nearly constant as compared to the healthy case; this is due to the increase in the number of conducting semiconductor devices during the post-fault. Here, due to an increase in the power loss the converter's efficiency decreases as shown in Fig. 17 (a) and Fig. 17 (b). However, it is noted from the figure that the little increase in the power loss results in a very slight change in efficiency.

Furthermore, the % current and voltage THD of the proposed converter is obtained under healthy and different fault combinations in MATLAB software and tabulated in Table 6. From the table, it is clear that the % THD of load current and voltage under healthy conditions, and fault cases reconfigured to 5 levels give the same value of 1.21 and 33.97, respectively. However, when fault cases are reconfigured to 3 levels, the % THD of load current and voltage increases to 2.24 and 66.42, respectively as shown in Fig. 17 (c).

6. Reliability analysis

Reliability is a crucial factor for a fault-tolerant converter, as it provides information about the estimated duration of the converter's usability. In this context, the Markov chain model [43] is utilized to determine the approximate lifespan of the converter with and without the redundant unit. If the Markov chain model considers three states (*A*, *B*, and *C*) as shown in Fig. 18, where state '*A*' represents a healthy condition, '*B*' indicates a faulty state with rated power, and '*C*' signifies a failure state, then the reliability function can be expressed as given below[42]-[43]:

$$R(t) = \frac{\lambda_{AC} - \lambda_{BB}}{\lambda_{AA} - \lambda_{BB}} e^{-\lambda_{AA} t} + \frac{\lambda_{AB}}{\lambda_{AA} - \lambda_{BB}} e^{-\lambda_{BB} t} \tag{12}$$

where, $\lambda_{AA}=\lambda_{AB}+\lambda_{AC}$, $\lambda_{BB}=\lambda_{BC}$, λ_{AB} represents the transition failure rate between A and B, λ_{BC} denotes the transition failure rate between *B* and *C*, and λ_{AC} indicates the transition failure rate between *A* and *C*.

The transition failure rate is dependent on the failure rate of switches, which is calculated according to the Military Handbook MIL-217F [43,44] and detailed in Table 7. Further, the transition failure

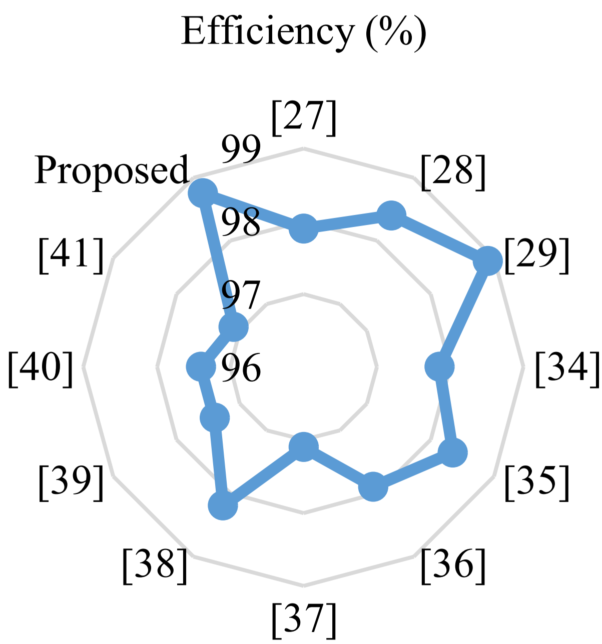


Fig. 22. Efficiency % of the literature and proposed converters.

rates between A and B, B and C, and A and C, with and without the redundant unit, are obtained and provided in Table 8.

Using Equation (12), Table 7, and Table 8, the reliability functions are obtained as given below:

$$R(t) = 0.969e^{-80.833t} + 0.030e^{-10.65t} \tag{13}$$

$$R(t) = 0.171e^{-80.833t} + 0.829e^{-5.171t} \tag{14}$$

Now, using equations (13) and (14) given above, the reliability curves are plotted against time, as shown in Fig. 19. The proposed converter demonstrates significantly better reliability with redundant switches compared to without them. Furthermore, the mean-time-to-failure (MTTF) value of the proposed converter is calculated as per [43]. The MTTF values with and without redundant switches are 0.0148×10^{-6} and, 0.1624×10^{-6} Respectively. This indicates that the proposed converter with redundant switches exhibits a more than ten times improvement in MTTF compared to the converter without redundant switches.

7. Comparative analysis

This section presents a comprehensive comparison of recent five-level fault-tolerant converters in the context of different parameters as tabulated in Table 9. The parameters considered for the comparison are: DC sources, number of main MOSFET/IGBT switches, number of redundant MOSFET/IGBT switches, number of trials, number of clamping diode, number of DC-link/flying capacitors, the total number of semiconductor devices, total standing voltage, total switch device power rating, efficiency, and multiple switch fault-tolerant ability.

Referring to Table 9, the works, [34,36], and [38–40] are flying capacitor-based MLIs and hence use a single DC source and one flying capacitor. The work [41] is also a flying capacitor-based MLI but uses one extra capacitor compared to the other flying capacitor-based MLIs. While, the remaining works employ two DC sources for generating five-level output voltages with the exception of [29], which uses 4 DC sources. The proposed converter, [29] and [38] employ a minimum number of main MOSFET/IGBT switches. Further, the converters [28] and [29] use a minimum number of the total semiconductor devices, followed by the proposed converter, [34 35] and [38]. But rated output reconfiguration is not possible in [27,28], and [29]. In [34]-[41], rated output is possible during the post-fault but all employ parallel connected redundant switches making concern for fault probability.

7.1. Total standing voltage

For a converter having a total k number of switches, total standing voltage (TSV) can be given in (15). where, V_{S_1} is the blocking voltage of the switch S_1.

$$\sum_{i}^{k} V_{S_i} \tag{15}$$

Using the above equation TSV of the literature and proposed converters are obtained as depicted in Fig. 20, and given in Table 9.

Referring to Fig. 20, the converters [37] and [39–41] have the highest TSV values, and the converters [26–29] and proposed have the lower TSV values. The lower value of the TSV not only indicates better efficiency but also indicates better reliability.

7.2. Total switch device power rating

For a converter having a total k number of switches, the total switch device power (SDP) rating can be given in (15). where, V_{S_1} is the blocking voltage of the switch S_1, I_{S_1} is the current stress of the switch S_1.

$$\sum_{i}^{k} V_{S_i} \times I_{S_i} \tag{16}$$

Using the above equation, the SDP rating of the literature and proposed converters are evaluated as depicted in Fig. 21 and given in Table 9.

Referring to Fig. 21, the converter [29] has the minimum SDP rating which is obtained by considering $V_{dc} = 200$ V and $R = 100$ O (i.e., $I_{load} =$ 2A). The proposed converters have comparable lower SDP values to converters [27] and [34–36]. The lower value of the SDP rating value indicates the cost-effectiveness of a converter.

7.3. Efficiency

As discussed in section 6, the efficiency of the literature and proposed converters are obtained using equation (8). Here, for calculation of the converter's efficiency, a 1 kW rating is considered and efficiency is calculated under the healthy condition as depicted in Fig. 22 and given in Table 9.

It is evident from Fig. 22 that the proposed converter and [29] exhibit slightly higher efficiency than the other converters, wherein [41] has the lowest efficiency.

8. Conclusion

This paper introduced a robust, fault-tolerant multilevel inverter designed to maintain an uninterrupted power supply even under fault conditions. By integrating two additional redundant switches, the proposed inverter architecture achieves substantial improvements in resilience, effectively handling both single and multiple switch faults with minimal impact on performance. Furthermore, this study addresses the issue of isolated DC source voltage fluctuations by modifying the LSPWM technique and optimizing it in response to variations in DC source voltages. Rigorous simulations validate the proposed converter's performance, while experimental tests on a 500 W prototype affirm its practical reliability and robustness. Comparative analyses with leading-edge converters underscore the benefits of the proposed design, such as reduced Total Switching Voltage (TSV), lower Static Device Power (SDP) ratings, improved efficiency, and advanced fault tolerance. Collectively, these features demonstrate the superior efficacy and reliability of the proposed inverter in high-demand applications, establishing it as a valuable advancement in power electronics.

CRediT authorship contribution statement

Vemana Ramanarayana: . **Kudithi Nageswara Rao:** . **Balram Kumar:** Writing – review & editing, Validation, Formal analysis. **S.V.K. Naresh:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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